

Homework Problems. *Solutions should be written/typed in numerical order and clearly labeled. Each page should have two problems (except the last page if there are an odd number of problems). Unsolved problems should be included but left blank.*

1. Write in tabular form:

a. $A = \{x \mid x \text{ is a letter in "correct"}\}$

$$A = \{c, o, r, e, t\}$$

b. $B = \{x \in \mathbb{R} \mid x \text{ is positive and } x \text{ is negative}\}$

$$B = \emptyset \text{ since no number is negative and positive. } 0 \text{ is not negative or positive.}$$

c. $C = \{x \in \mathbb{Z} \mid x - 2 \leq 5\}$

$$C = \{\dots, 4, 5, 6, 7\}$$

2. Write these sets in a set-builder form:

a. Let $A = \{3, 4, 5, 6, 7, \dots\}$

$$A = \{x \mid x \in \mathbb{Z} \text{ and } x \geq 3\}$$

b. Let $B = \{3, 5, 7, 9, \dots\}$

$$B = \{x \in \mathbb{Z} \mid x \geq 3 \text{ and } x \text{ is odd}\}$$

c. Let C be the US presidents elected after 1999.

$$C = \{x \mid x \text{ is a US president elected after 1999}\}$$

3. Find all the subsets of $A = \{x, y, z\}$.

$$\{x\} \quad \{x, y\} \quad \{x, y, z\}$$

$$\{y\} \quad \{x, z\} \quad \{\emptyset\}$$

$$\{z\} \quad \{y, z\}$$

4. Prove that every nonempty subset has a proper subset.

Proof. Let $A \neq \emptyset$. $\emptyset \subseteq A$ and $A \neq \emptyset$. So then $\emptyset$ is a proper subset of A , i.e., $\emptyset \subset A$. □

5. Prove that if A is a subset of $\emptyset$, then $A = \emptyset$.

Proof. $\emptyset \subseteq A$ since every element in $\emptyset$ is also in A , and by assumption $A \subseteq \emptyset$ by the hypothesis. So then $A = \emptyset$. □

6. Prove that $A = \{2, 3, 4, 5\}$ is not a subset of $B = \{x \mid x \text{ is even}\}$.

Proof. 5 is not even since $5 = 2 \cdot 2 + 1$ (2 does not evenly divide 5). So then $5 \notin B$ which implies that $A \not\subseteq B$ since $5 \in A$. □

7. Prove that A and B are subsets of $A \cup B$.

Proof. Let $a \in A$. Since $a \in A$, a is an element of A or B . Thus $a \in A \cup B$ which implies that $A \subseteq A \cup B$.

Similarly, we can show that $B \subseteq A \cup B$. □

8. Prove that $A = A \cup A$ and $A = A \cap A$.

Proof. ($A = A \cup A$). By exercise 7, $A \subseteq A \cup A$. So we need only show that $A \cup A \subseteq A$. Let $a \in A \cup A$. Then $a \in A$ or $a \in A$, implying that $A \cup A \subseteq A$. Thus $A \cup A = A$.

($A = A \cap A$). Let $a \in A$. Then $a \in A$ and $a \in A$. Thus $A \subseteq A \cap A$. Let $a \in A \cap A$. Then $a \in A$ and $a \in A$ by definition, and this implies that $A \cap A \subseteq A$.

Thus $A = A \cap A$. □

9. Prove that $A \cap B$ is a subset of A and of B .

Proof. Let $x \in A \cap B$. Then $x \in A$ and $x \in B$ by definition. So then $x \in A$ implying that $A \subseteq A \cap B$. Similarly we can show that $B \subseteq A \cap B$. $\square$

10. Prove that $A \cup B = \emptyset$ implies that $A = \emptyset$ and $B = \emptyset$.

Proof. Suppose that $A \cup B = \emptyset$. $A \subseteq A \cup B = \emptyset$ by exercise 7. Thus $A \subseteq \emptyset$, and this implies that $A = \emptyset$ by exercise 5.

Similarly we can show that $B = \emptyset$. $\square$

Presentation problems.

1. Prove that $A - B$ is a subset of $A \cup B$.
2. Prove that if A a subset of B then $A \cap B = A$.
3. Let $A \cap B = \emptyset$. Prove that $B \cap A' = B$.
4. Prove that if A is a subset of B then $A \cup B = B$.
5. Prove that $A' - B' = B - A$.
6. Prove that if A is a subset of B then B' is a subset of A' .
7. Prove that if $A \cap B = \emptyset$ then $A \cup B' = B'$.
8. Prove that $(A \cap B)' = A' \cup B'$.
9. Prove that $A \subset B$ implies that $A \cup (B - A) = B$.